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Pallet sled and delivery system

Abstract

A pallet sled includes a pair of tines extending forward from a base. A load wheel supports each of the pair of tines. The load wheels are each configured to move away from the respective tine to lift the respective tine upward relative to a support surface on which the load wheel is supported. A processor on the pallet sled records usage of the sled, such as lift cycles, location over time, battery condition, lift height, weight lifted, distance traveled, and the like. A communication circuit may be configured to send data from the pallet sled to a remote server. An identification reader may be configured to read an identification tag of an object supported on the pair of tines, such as an rfid tag on a pallet.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3576333	12/1970	Danielson	280/43.12	B62B 3/0618
4061237	12/1976	Austin	414/592	B66F 9/07563
6669089	12/2002	Cybulski	235/383	B66F 9/0755
6855894	12/2004	Van Seumeren	177/139	B62B 3/0606
7401054	12/2007	Shah	707/999.103	G06F 16/289
7974925	12/2010	Shah	709/229	G06F 16/289
7976032	12/2010	Dockins	187/229	B62B 3/0618
7979355	12/2010	Shah	709/229	G06F 16/289
8979099	12/2014	Ellis	280/43.12	B66F 9/142
10336543	12/2018	Sills	N/A	G05D 1/0094
11068841	12/2020	Lui	N/A	G06K 7/10376
11279569	12/2021	Theobald	N/A	B65G 47/912
11840271	12/2022	Martin, Jr. et al.	N/A	N/A
2005/0200457	12/2004	Bridgelall	340/10.3	G06K 7/10346
2006/0181039	12/2005	Fridlington, Jr.	280/43.12	B62B 3/0618
2006/0231312	12/2005	Passeri	280/47.11	B62B 3/0612
2006/0255949	12/2005	Roeder	340/572.7	G06K 7/10079
2008/0044265	12/2007	Borntrager	211/1.51	B66F 9/065
2008/0243873	12/2007	Shah	N/A	G06F 21/6227
2008/0243915	12/2007	Shah	N/A	G06F 16/289
2009/0059004	12/2008	Bochicchio	348/148	B66F 17/003
2009/0115966	12/2008	Waldorf	351/210	A61B 3/113
2009/0183953	12/2008	Ellington	187/231	B62B 3/0606
2009/0240554	12/2008	Oswald	348/207.1	G06Q 50/26
2009/0303035	12/2008	Kirtley, Jr.	340/539.11	G08B 13/24
2012/0223812	12/2011	Kail	340/10.1	B66F 9/085
2012/0269606	12/2011	Newell	414/495	B62B 3/0612
2014/0291594	12/2013	Newell	254/2R	B62B 5/0033

2016/0376135	12/2015	Kim	701/50	B66F 9/0755
2017/0011364	12/2016	Whitman	N/A	H04L 67/01
2017/0325675	12/2016	Liu	N/A	A61B 3/0025
2017/0349197	12/2016	Jonasson	N/A	B66F 9/063
2018/0029797	12/2017	Hance	N/A	B65G 1/0492
2018/0114415	12/2017	Mattingly	N/A	G08B 21/182
2018/0121864	12/2017	Sullivan	N/A	B65G 67/02
2020/0002143	12/2019	Hasegawa	N/A	G05D 1/0246
2020/0269896	12/2019	Kalinowski	N/A	B62B 3/0618
2020/0327497	12/2019	Marvizadeh	N/A	G06Q 10/08355
2021/0272037	12/2020	Hanebeck	N/A	G06Q 10/08
2021/0403066	12/2020	Martin, Jr.	N/A	G05B 19/4155
2021/0403103	12/2020	Alshaalan	N/A	B66F 9/07554

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2019182560	12/2018	JP	N/A

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Background/Summary

BACKGROUND

(1) Many items are delivered to stores on pallets. A plurality of loaded pallets may be loaded onto a truck. One or more of the pallets may be destined for each of a plurality of stores. At each store, the driver uses a pallet sled (which may be a pallet jack or the like) to lift one or more pallets and bring them into the store. In the store, the pallets may be unloaded. The empty pallets are then returned to the truck on the pallet sled.

SUMMARY

(2) A pallet sled includes a pair of tines extending forward from a base. A load wheel supports each of the pair of tines. The load wheels are each configured to move away from the respective tine to lift the respective tine upward relative to a support surface on which the load wheel is supported. A processor on the pallet sled records usage of the sled, such as lift cycles, location over time, battery condition, lift height, weight lifted, distance traveled, and the like. A communication circuit may be configured to send data from the pallet sled to a remote server. An identification reader may be configured to read an identification tag of an object supported on the pair of tines, such as an rfid tag on a pallet.

(3) The pallet sled may further include a gps receiver mounted on the sled, the communication circuit configured to receive a location of the pallet sled from the gps receiver and transmit the location to the remote server.

(4) The pallet sled may include an rfid reader configured to read an rfid tag of a pallet on the tines of the pallet sled. The pallet sled may include a processor programmed to analyze the rfid tag of the pallet on the tines to determine if the pallet on the tines should be delivered. The pallet sled may include a gps receiver mounted on the sled wherein the processor is programmed to determine if the pallet on the tines should be delivered based upon a current location as determined by the gps receiver.

(5) The pallet sled may include a weight sensor configured to weigh objects on the pair of tines.

The processor may be programmed to analyze a weight of the pallet on the tines and to determine based upon a weight of the pallet if the pallet is loaded correctly.

(6) The processor may be programmed to monitor usage of the pallet sled and to communicate data regarding the usage of the pallet sled to the remote server. The usage of the sled that the processor is programmed to monitor may include time at each store to which the pallet sled is used to deliver items. The usage of the sled that the processor is programmed to monitor may include unloading time in a store.

(7) The processor may be programmed to monitor number of cycles that the pair of tines are raised to a lift position.

(8) A method for making a delivery using the pallet sled may include the step of reading an identification tag of a first pallet of a plurality of pallets with the identification reader. Each of the plurality of pallets loaded with a plurality of items. Each of the plurality of pallets has an identification tag associating said each pallet with a different one of a plurality of stores including a first store. The method further includes determining a current location of the plurality of pallets to be at the first store. Based upon the current location and the identification tag, it is determined if the identification tag of the first pallet is associated with the first store. Based upon a determination that the pallet id of the first pallet is associated with the first store), the first pallet is delivered to the first store with the pallet sled.

(9) A method for operating a pallet sled may include the step of bringing the pallet sled and a plurality of pallets to a first store in a truck, each of the plurality of pallets having at least one item stacked thereon. At least one tine of the pallet sled is positioned below a first pallet of the plurality of pallets. The at least one tine of the pallet sled is raised to lift the first pallet. The first pallet is brought from the truck to the first store on the at least one tine of the pallet sled. The at least one tine of the pallet sled may be lowered in the store to lower the first pallet onto a floor in the store. The at least one item on the first pallet is unloaded in the store.

(10) The first pallet with the at least one tine is lifted in the store. The first pallet is brought from the store to the truck on the pallet sled. At least one of distance, weight, or cycles of the pallet sled is monitored during the forgoing steps.

(11) The data gathered may be recorded on the pallet sled. The data gathered during said step i) may be transmitted from the pallet sled to a remote server.

(12) A pallet sled includes a base and a pair of tines extending forward from the base. A load wheel supports each of the pair of tines. The load wheels are each configured to move away from the respective tine to lift the respective tine upward relative to a support surface. A processor recording data reflective of usage of the pallet sled. The processor may also record cycles of the pair of tines being lifted. The processor may record a location of the pallet sled over time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic view of a delivery system.

(2) FIG. 2 is a schematic of the pallet sled of FIG. 1 according to one embodiment.

(3) FIG. 3 is a perspective view of the pallet sled of FIG. 2.

(4) FIG. 4 is a perspective view of the underside of the pallet sled of FIG. 1.

(5) FIG. 5 shows the pallet sled of FIG. 1 with the tines in the spaced apart position.

(6) FIG. 6 shows one of the tines of FIG. 3, partially broken away.

(7) FIG. 7 shows the tine of FIG. 6 raised to the lift position.

(8) FIG. 8 shows the tine of FIG. 6 in the side shifting position.

(9) FIG. 9 shows an example display screen on the server of FIG. 1.

DETAILED DESCRIPTION

(10) FIG. 1 is a high-level view of a delivery system 8 including one or more distribution centers 32, a central server 34 (e.g. cloud computer), and a plurality of stores 36. A plurality of trucks 38 or other delivery vehicles each transport a plurality of the items 40 on pallets 42 from one of the distribution centers 32 to a plurality of stores 36. A wheeled pallet sled 10 is on each truck 38 to facilitate delivery of one or more pallets 42 of goods 40 to each store 36.

(11) At each store 36 the driver's mobile device 50 indicates which of the loaded pallets 42 (based upon their pallet ids) are to be delivered to that store 36 as verified by gps on the mobile device 50. The driver chooses the correct pallet(s) 42 for that location by reading the pallet id (rfid, barcode, etc) with the mobile device 50. The driver moves one or more of the loaded pallets 42 into the store 36 at a time with the pallet sled 10.

(12) FIG. 2 is a schematic of the pallet sled 10. The pallet sled 10 can include an identification reader, such as an rfid reader 90, to check the pallet identification of pallet 42 carried thereon by reading the identification tag, such as rfid tag 94, secured to the pallet 42. The pallet 42 may be a half pallet as shown or a full pallet. The rfid reader 90 may also read rfid tags 96 on the items 40 (if so provided). The rfid reader 90 and rfid tag 94 can be active, passive or a mixture of any radio frequency identification systems. Alternatively, the identification reader and identification tag could utilize other electromagnetic signals to provide a pallet id from the id tag to the id reader, such as optical, including bar codes, qr codes, or the like.

(13) Optionally the pallet sled 10 may include a camera 92 for imaging the loaded pallet 42 carried thereon for validation. In the configuration shown, the camera 92 would only image the short side of the loaded pallet 42 (and the top of the loaded pallet 42).

(14) A GPS receiver 91 ("GPS" referring generically to any global navigation satellite system) or other location-determining device may be mounted to the pallet sled 10. A CPU 93 (with associated memory, storage, processor, suitable programming, etc and optionally a touch screen and keyboard and mouse) can also be mounted to the pallet sled 10 and is connected to the other devices on the pallet sled 10.

(15) A local wireless communication circuit 95 (e.g. Bluetooth) is also connected to the CPU 93. The communication circuit 95 may communicate the pallet id of the pallet 42 on the pallet sled 10 to the driver's mobile device 50 and/or to the CPU 93. A weight sensor 97 may be integrated into the pallet sled 10 configured to measure weight on the tines to provide a weight of the loaded pallet 42 to the CPU 93 and/or mobile device 50.

(16) A cell data circuit 99 could also be included on the pallet sled 10 and could be used to communicate directly with the central server 34 (FIG. 1), including sending and receiving data and commands. Alternatively, the local communication circuit 95 could provide that same communication by accessing the cell data circuit on the driver's mobile device 50 (e.g. "tethering").

(17) The pallet sled 10 could also include a temperature sensor 100. The temperature sensor 100 could monitor ambient air temperature or specifically a temperature of a battery that powers the pallet sled 10. Optionally, two temperature sensors could monitor both ambient air and battery temperature. The temperature sensor(s) 100 report measured temperature to the CPU 93 when polled or periodically.

(18) Much of the hardware on the pallet sled 10 including the CPU 93, GPS receiver 91, camera 92, communication circuit 95, and cell data circuit 99 could all be provided in the form of a mobile device (e.g. smartphone or tablet) mounted to the pallet sled 10. The CPU 93 is connected to all of the hardware and sensors on the sled 10. The CPU 93 also controls actuation of the tines, as directed by the user. For example, the CPU 93 controls the actuator for lifting the tines and, optionally, an actuator for adjusting the tines toward one another.

(19) Optionally, all data is stored locally on CPU 93 and retrieved by a computer at the DC 32 via the local communication circuit 95 whenever the pallet sled 10 returns to the DC 32. The CPU 93 processes event triggers, such as delivery of a pallet 42. Alternatively, the data is sent directly to the server 34 by the CPU 93 via the communication circuit 95.

(20) In use, at each store **36**, the driver selects one or more of the loaded pallets **42** intended for that store **36** and loads it on the pallet sled **10** in the truck **38**. Based upon reading the rfid tag **94** on the pallet **42** and the current location from the gps receiver **91**, the pallet sled **10** determines whether the pallet **42** on the pallet sled **10** is assigned to the store **36** at the current location by accessing information from the server **34** from locally-stored list associating pallet ids and stores **36**. The pallet sled **10** may indicate a correct pallet **42** or an incorrect pallet **42** with a display (or other visual indicator) and/or sound on the pallet sled **10** and/or a notification sent to the driver's mobile device **50**.

(21) The camera **92** on the pallet sled **10** may take one or more images and send them to the server **34**. The server **34** analyzes the image(s) to confirm the presence of the correct number of items **40** of each of the SKUs associated with the pallet id of that pallet **42**, and then communicates a confirmation to the pallet sled **10**, the driver's mobile device **50** and/or the store employee's mobile device, which is displayed on the screens. The server **34** may use machine learning or other image analysis techniques to identify the SKUs on the pallet **42**.

(22) If the pallet sled **10** weighs the loaded pallet **42**, this information can also be used to validate the loaded pallet **42**. The CPU **93** (or server **34**) compares the weight from weight sensor **97** to an expected weight of the empty pallet **42** plus the expected weight of all the SKUs that should be loaded on the pallet **42**. Alternatively, the CPU **93** (or server **34**) can compare the weight from the weight sensor **97** to an actual weight of the loaded pallet **42** that was measured at the DC **32** after it was loaded. This verifies that nothing has changed (added or removed) since the pallet **42** was loaded and verified at the DC **32**.

(23) If the pallet sled **10** only images the short side of the loaded pallet **42**, then a complete validation of the items **40** on the loaded pallet **42** cannot be done, but it can be enough of a confirmation that the items **40** on the loaded pallet **42** have not been disturbed since leaving the distribution center, especially if the loaded pallet **42** is wrapped and especially again if the loaded pallet **42** has the right weight.

(24) If a discrepancy is detected, the CPU **93** indicates the discrepancy to the driver (either via an interface on the CPU **93** or via the driver's mobile device **50**). There may only be a general indication of a discrepancy if the system does not have enough information (e.g. only a short side image), or the system may be able to indicate the specific discrepancy and how to remedy the discrepancy to the driver.

(25) If a general discrepancy is indicated, the driver can then use the driver's mobile device **50** to take a plurality of images of the loaded pallet **42** (or possibly a single image from a corner). These images can be uploaded to the server **34** to be processed e.g. via machine learning to identify specific SKUs on the loaded pallet **42** and to identify a specific discrepancy for the driver to correct (e.g. one or more missing SKUs and/or extra SKUs).

(26) The driver can correct the discrepancy by retrieving items **40** of the missing SKUs from the truck **38** or crediting the missing SKUs to the store account **42**. Any SKUs detected that do not belong on the pallets **42** can be returned by the driver to the truck **38**. On the store worker's mobile device **52** (via an app), the store worker confirms the presence of the loaded pallet **42** and receives a list of SKUs associated with that pallet id from the server **34**.

(27) The pallet sled **10** can also assist in tracking the return of the pallets **42** and returnable packaging such as plastic beverage crates **98**. The pallet sled **10** can count the number of crates **98** and the pallets **42** that are being returned to the truck **38** and report those numbers to the server **34**. If the returnable packaging, such as plastic beverage crates **98**, have rfid tags **96**, the rfid reader **90** on the pallet sled **10** can count the number of crates **98** that are being returned to the truck **38**. The camera **92** and/or weight sensor **97** can also estimate the type and quantity of returnable packaging and/or pallets **42** that is being returned. Over time, this can provide asset tracking information. For example, this enables the server **34** to determine if the number of pallets **42** and/or number of crates **98** delivered to a particular store **36** consistently exceeds the number of pallets **42** and/or number of

crates **98** that are returned from that store **36**, thus indicating that the store **36** is experiencing a high rate of asset loss for some reason, which can then be investigated and remedied.

(28) The CPU **93** tracks the location of the pallet sled **10** via the GPS receiver **91** over time. The CPU **93** tracks when the pallet sled **10** lifts a pallet **42** (because the CPU **93** activates the actuator to lift the tines), the weight lifted by the pallet sled **10** each time, which pallet **42** is lifted by the pallet sled **10** (and when), and/or which pallet **42** is carried by the pallet sled **10** into which store **36** (and when). This information is transmitted to server **34**.

(29) The CPU **93** monitors and accumulates biometric data of the pallet sled **10**, such as time in the truck **38**, time walking from the truck **38** to the store **36**, time for check-in, time for stocking shelves and/or, time for check out, time walking from store to truck. Time in the truck can be determined from the GPS receiver **91**. Time walking from the truck **38** to the store **36** can be determined from the GPS receiver **91**. Time for check-in can be determined as time after the pallet sled **10** is in the store **36** before pallet unloading begins, as determined by the camera **92** and/or the weight sensor **97** detecting removal of the items **40** and/or the actuator lowering the tines. Time for stocking shelves can be determined as time after the pallet sled **10** is in the store **36** after pallet unloading begins, as determined by the camera **92** and/or the weight sensor **97** detecting removal of the items **40**, until pallet **42** is determined to be empty by the camera **92** and/or weight sensor **97**. Time for check out can be determined to be time after the pallet **42** is empty until pallet sled **10** is moved out of the store **36**, as determined by GPS receiver **91** and/or camera **92**. Time walking from the store to the truck **38** can be determined by the GPS receiver **91** and/or camera **92**.

(30) This data is all sent to the server **34** (FIG. 1) and compiled for the plurality of pallet sleds **10**. With this data, the server **34** determines the time spent for each stop (each store) and for each route. The server **34** determines the time spent by customer and by franchise. The server **34** reconciles assets (pallets **42** and returnable packaging, such as crates **98**), as explained above. The server **34** determines seasonality trends, performs drive and route comparisons, performs region metrics comparisons, and provides industry comparisons.

(31) The CPU **93** collects data regarding usage of the pallet sled **10** from multiple sensors on the pallet sled **10**, such as one or more of the following: a sensor detecting lift cycles (how many times the pallet sled **10** has lifted a pallet **42**) or the actuations of the lift actuator commanded by the CPU **93**, the weight sensor **97** (how much weight is being lifted by the pallet sled **10** and how much weight is on the pallet sled **10** as it is being rolled a measured distance across parking lots and store floors), how much distance has the pallet sled **10** been rolled while empty and while loaded, average and extreme temperatures experienced by the pallet sled **10** during use. If the pallet sled **10** has expandable tines, the CPU **93** collects cycles of the expansion and retraction of the expandable tines. This collected data is sent to the server **34** for preventative and predictive maintenance analysis.

(32) FIGS. 3-8 show one possible configuration of the mechanical features of the pallet sled **10**. Referring to FIG. 1, the pallet sled **10** includes a lift module or base **12** containing the lift mechanisms and the tine spacing actuators **15**. The base **12** may also contain all of the hardware shown schematically in FIG. 2, most of which could be provided in a mobile device (such as smart phone or tablet). The base **12** is connected to a pair of fork tines **14**. The base **12** can raise and lower the fork tines in a known manner, by leveraging the load wheels **16** supporting the tines **14**, such as by hydraulics, pneumatics, electric motors, or via a manual actuator. The base **12** can also move the tines toward and away from one another with the tine spacing actuators **15**, which could be hydraulics, pneumatics, electric motors driving a ball screw or a threaded rod, or via a manual actuator.

(33) The pallet jack mechanism could be semi-electric, e.g. electric drive system and manual lifting (via pumping the tiller arm) or full-electric, e.g. electric drive system and electric lifting/lowering. In this example, the lift is semi- or full-electric, so battery power exists on the equipment to drive the actuators whether it be an electric hydraulic pump or electric motor. Other power sources could

be provided, including manual.

(34) The pallet sled **10** may provide a first mechanism for facilitating the adjustment of the tine spacing without dragging the load wheels **16**.

(35) FIG. **4** is a perspective view of the underside of the tines **14** of FIG. **1**. Each tine **14** includes a load wheel **16** which is rotatable on an axis **18** perpendicular to the long axis of the tine **14**. The load wheel axis **18** is mounted at the end of a pair of arms **30** which are pivotable about an axis **20** fixed to the tine **14**. The axis **20** is spaced rearwardly of the axis **18**, and also perpendicular to the long axis of the tine **14**. A side wheel **22** is mounted to arms **23** extending forwardly of the load wheel **16** and is oriented in a direction such that its axis is transverse to the load wheel axis **18**. The arms **23** are pivotable about the axis **18** (but could alternatively pivot about an axis parallel to that load wheel axis **18**) in a vertical plane parallel to the long axis of the tine **14**.

(36) As explained below, the side wheel **22** can facilitate lateral motion of the tine **14**. A push rod **26** within each tine **14** pivots the arms **30** to deploy and retract the load wheels **16**, in a known manner.

(37) In FIGS. **3** and **4**, the tines **14** of the pallet sled **10** are spaced closed to one another. In FIG. **5**, the tines **14** are spaced further apart. The actuators **15** for moving the tines **14** toward and away from each other are shown on the base **12**. As will be explained below, the side wheels **22** permit the tines **14** to be moved toward and away from each other without dragging the load wheels **16** parallel to their axes **18**, which could cause damage to the load wheels **16** and/or the floor.

(38) FIG. **6** shows a side view of the tine **14** (without the base **12**, for simplicity) and with part of the tine **14** broken away for illustration. The tine **14** is in the low position with the load wheel **16** retracted into the tine **14** as much as possible. In this position, the tines **14** are closest to the floor and can be slid under a pallet. The load wheel **16** is rotatable on the axis **18** at the end of the arm **30** which pivots about an axis **20** that is fixed to the tine **14**. The push rod **26** is pivotably secured to a rearward portion of the arm **30** at a pivot point that is rearward and upward of the axis **20**. In a generally known manner, the push rod **26** can, by forward motion, cause the load wheel **16** to deploy downward and lift the tine **14**. By rearward motion, the push rod **26** causes the load wheel **16** to retract upward into the tine **14**, lowering the tine **14** relative to the floor. The arms **30** each include a stop **36** projecting upward from a point rearward of the load wheel axis **18**.

(39) Still referring to FIG. **6**, stretching the springs **33** permits the load wheels **16** to move closer to the underside of the tines **14** by permitting rotation of the arms **23** downward away from the arms **30** to the position where both the arms **23** and the arms **30** are substantially horizontal. In FIG. **7**, the springs **33** are contracted.

(40) FIG. **7** shows the tines **14** in a lifted or high position, such as would be used to lift a pallet off a floor and move the pallet. The arms **30** are pivoted downward, thereby moving the load wheels **16** away from the tines **14**, causing the tines **14** to move away from the floor, with the load wheels **16** on the floor. The side wheel **22** is rotatable on an axis almost parallel to arm **23**. A spring **33** biases the arm **23** upward rotationally toward the arm **30** and toward the tine **14**. The push rod **26** is moved forward to a mid position, such that the arms **30** pivot downward about axis **20** until the load wheel **16** is moved to a lift position away from the tine **14** but the side wheel **22** is still spaced above the floor. As the arms **30** pivot downward away from the underside of the tine **14**, the spring **33** is able to pull the arms **23** about the axis **18** upward toward the arms **30** (and toward the tine **14**) until the arms **23** contact the stops **36** on the arms **30**. In this manner, the springs **33** hold the side wheels **22** off the floor in this high position.

(41) FIG. **8** shows the tines **14** of the pallet sled **10** with the load wheels **16** and side wheels **22** moved to the side-shifting position, i.e. the side wheels **22** are in contact with the floor and the load wheels **16** are lifted off the floor. The push rod **26** has been moved forward to its furthest forward position, rotating the arms **30** downward to a maximum degree. This also rotates the arms **23** further downward until the side wheels **22** contact the floor, then lifting the load wheel **16** off the floor, and lifting the tines **14** even a little higher (this would not be done while the tines **14** are

loaded). Each side wheel **22** provides a support surface configured to selectively be the lowermost surface of the tine to facilitate lateral displacement of the first tine.

(42) The tines **14** can then be moved back toward one another as shown in FIGS. **3** and **4**. The tines **14** would roll on the side wheels **22**. With the load wheels **16** off the floor, movement of the tines **14** laterally does not damage the load wheels **16** or the floor. In this side-shifting position, the tines **14** can be moved laterally toward and away from one another. With the tines **14** closely-spaced near one another or spaced further apart from one another, the load wheels **16** and tines **14** can be moved back and forth between the lift position, the low position, and the side-shifting position.

(43) FIG. **9** shows an example display screen **110** from the server **34** of FIG. **1** (such as a workstation or mobile device remotely accessing data on the server **34**). The display screen **110** will only show information regarding pallet sleds **10** to which the user has permission. For example, a site manager will only have access to their pallet sleds **10**, while a regional manager will have access to all of the pallet sleds **10** in the area.

(44) A small status window **112** for each pallet sled **10** (FIG. **1**) is displayed on the screen **110**. Each status window **112** gives a summary of the status of the associated pallet sled **10**, which is identified in the window **112**, including for example, whether the pallet sled **10** is currently connected to the server **34** (e.g. via communication circuit **95**, FIG. **2**), the current charging level of the battery in the pallet sled **10**, whether there are any alerts associated with the pallet sled **10**, and the ambient temperature around the pallet sled **10**. The battery for powering the pallet sled **10** may only operate (charge or discharge) within a certain temperature range. Therefore, if the user sees a temperature outside that range, the battery may need to be removed for charging in a temperature-controlled environment.

(45) A map **114** shows a current location of the pallet sleds **10**, including whether there are associated alerts.

(46) If the user selected one of the windows **112**, the rest of the display shows more information. An alert window **116** is displayed for the selected pallet sled **10** giving more detail about the alerts, such as errors, failures, necessary maintenance, or other incidents. Location window **118** shows a specific location of the pallet sled **10**, such as the lat/long and the date/time that the lat/long was received by the server **34**. A time usage window **120** shows the elapsed time (e.g. hours) that the pallet sled **10** has been in an “on” mode and actual “use” time.

(47) A cycle window **122** displays the number of lift cycles and (if applicable) the number of expansion/retraction cycles. A lift distance window **124** displays the total lift distance by the tines of the pallet sled **10** both in terms of a total for the current day and lifetime total. A service window **126** displays the key-on time since the last service on the pallet sled **10** and the key-on time until the next service should be performed. This time until next service could be based upon actual usage of the pallet sled **10**, i.e. it could be based upon key-on time, use time, lift cycles, expansion/retraction cycles, and/or lift distance, or any combination or weighted combination of these. In the service window **126** specific service items may also be specified based upon reports from sensors onboard the pallet sled **10** and/or directly from the user.

(48) Expanding the service window **126** (expanded window not shown) could provide a specific service alert regarding that pallet sled **10**. For example, there may be a message describing the problem with the pallet sled **10** along with a recommendation for troubleshooting and/or repairing the problem. The display may provide a link that will order the necessary parts of the pallet sled **10** (based upon the detected error) and the specific steps needed to diagnose the problem further and/or to make the repair, such as how to install a replacement part. The service window **126** could alternatively be displayed by the CPU **93** on the pallet sled **10** itself, or on the mobile device **50**.

(49) In accordance with the provisions of the patent statutes and jurisprudence, exemplary configurations described above are considered to represent preferred embodiments of the inventions. However, it should be noted that the inventions can be practiced otherwise than as specifically illustrated and described without departing from its spirit or scope. Alphanumeric

identifiers on method steps are solely for ease in reference in dependent claims and such identifiers by themselves do not signify a required sequence of performance, unless otherwise explicitly specified.

Claims

1. A pallet sled comprising: a base; a pair of tines extending forward from the base; a weight sensor configured to weigh objects on the pair of tines; a load wheel supporting each of the pair of tines, the load wheels each configured to move away from the respective tine to lift the respective tine upward relative to a support surface; a communication circuit configured to send data from the pallet sled to a remote server; and at least one processor programmed to monitor usage of the pallet sled, including weight lifted by the pallet sled each time the pair of tines are lifted, and to communicate data regarding the usage of the pallet sled, including weight lifted by the pallet sled each time the pair of tines are lifted, to the remote server via the communication circuit.
2. The pallet sled of claim 1 further including a gps receiver mounted on the pallet sled, the communication circuit configured to receive a location of the pallet sled from the gps receiver and transmit the location to the remote server.
3. The pallet sled of claim 1 further including an rfid reader configured to read an rfid tag of a pallet on the tines of the pallet sled.
4. The pallet sled of claim 1 wherein the at least one processor is programmed to analyze a weight of a pallet on the tines as determined by the weight sensor and to determine based upon a weight of the pallet if the pallet is loaded correctly.
5. The pallet sled of claim 1 wherein usage of the pallet sled that the at least one processor is programmed to monitor includes time at each store to which the pallet sled is used to deliver items.
6. The pallet sled of claim 1 wherein usage of the pallet sled that the at least one processor is programmed to monitor includes unloading time in a store.
7. The pallet sled of claim 1 wherein the at least one processor is programmed to perform maintenance analysis of the pallet sled based upon the monitored usage of the pallet sled including weight lifted by the pallet sled each time the pair of tines are lifted.
8. A pallet sled comprising: a base; a pair of tines extending forward from the base; a load wheel supporting each of the pair of tines, the load wheels each configured to move away from the respective tine to lift the respective tine upward relative to a support surface; a communication circuit configured to send data from the pallet sled to a remote server; and a processor programmed to monitor usage of the pallet sled and to communicate data regarding the usage of the pallet sled to the remote server via the communication circuit; wherein usage of the pallet sled that the processor is programmed to monitor includes number of cycles that the pair of tines are raised to a lift position and unloading time in a store.
9. A delivery sled comprising: a base; a pair of tines extending forward from the base for supporting an object thereon; a load wheel supporting each of the pair of tines, the load wheels each configured to move away from the respective tine to lift the respective tine upward relative to a support surface on which the load wheel is supported; a gps receiver determining a location of the delivery sled; and a processor configured to determine if the object should be delivered based upon the location of the delivery sled.
10. The delivery sled of claim 9 further including a communication circuit configured to send data from the delivery sled to a remote server.
11. The delivery sled of claim 9 wherein the processor is configured to determine that the object should be delivered to a first location and to determine that the object should be delivered based upon the location of the delivery sled as determined by the gps receiver being the first location.
12. A method for making a delivery using the delivery sled of claim 9 including: a) reading an identification tag of a first pallet of a plurality of pallets, each of the plurality of pallets loaded with

a plurality of items, each of the plurality of pallets having an identification tag associating said each pallet with a different one of a plurality of stores, including the first pallet being associated with a first store of the plurality of stores; b) the processor determining a current location of the plurality of pallets to be at the first store; c) the processor determining if the identification tag of the first pallet is associated with the first store; and d) based upon a determination that the identification tag of the first pallet is associated with the first store in step c), delivering the first pallet to the first store with the delivery sled.

13. A pallet sled comprising: a base; a pair of tines extending forward from the base; a load wheel supporting each of the pair of tines, the load wheels each configured to move away from the respective tine to lift the respective tine upward relative to a support surface; at least one processor recording data reflective of usage of the pallet sled, wherein the at least one processor is configured to record the data reflective of usage of the pallet sled including at least one of distance, weight, elapsed time, or temperature during usage of the pallet sled; and a communication circuit configured to send the data from the pallet sled to a remote server.

14. The pallet sled of claim 13 wherein the at least one processor records a location of the pallet sled over time.

15. The pallet sled of claim 14 wherein the data includes the location of the pallet sled.

16. The pallet sled of claim 13 wherein the at least one processor is configured to record distance traveled by the pallet sled during usage of the pallet sled.

17. The pallet sled of claim 13 wherein the at least one processor is configured to record the weight lifted by the pallet sled during usage of the pallet sled for a plurality of lift cycles.

18. The pallet sled of claim 13 wherein the at least one processor is programmed to perform maintenance analysis of the pallet sled based upon the data reflective of usage of the pallet sled.

19. The pallet sled of claim 13 in combination with the remote server, wherein the remote server is programmed to perform maintenance analysis of the pallet sled based upon the data reflective of usage of the pallet sled.

20. The combination of claim 19 wherein the remote server is programmed to perform maintenance analysis of the pallet sled based upon the data reflective of usage of the pallet sled including weight lifted by the pallet sled.

21. A pallet sled comprising: a base; a pair of tines extending forward from the base; a load wheel supporting each of the pair of tines, the load wheels each configured to move away from the respective tine to lift the respective tine upward relative to a support surface; at least one processor recording data reflective of usage of the pallet sled; and a communication circuit configured to send the data from the pallet sled to a remote server, wherein the at least one processor is configured to determine a quantity of returnable packaging or pallets that is being returned from a store on the pallet sled and to communicate the quantity via the communication circuit.

22. The pallet sled of claim 21 in combination with a server configured to receive and accumulate the quantity of returnable packaging or pallets that is returned from the store over time and to compare the quantity returnable packaging or pallets that is returned from the store to a quantity of returnable packaging or pallets that is delivered to the store to determine a rate of asset loss.

23. The pallet sled of claim 21 further including a camera mounted to the pallet sled, wherein the at least one processor is configured to use the camera to determine quantity of returnable packaging or pallets that is being returned from a store on the pallet sled.

24. The pallet sled of claim 21 further including a weight sensor configured to weigh objects on the pair of tines, wherein the at least one processor is configured to use the weight sensor to determine quantity of returnable packaging or pallets that is being returned from a store on the pallet sled.

25. The pallet sled of claim 24 further including a camera mounted to the pallet sled, wherein the at least one processor is configured to use the camera to determine quantity of returnable packaging or pallets that is being returned from a store on the pallet sled.

26. The pallet sled of claim 24 in combination with a server configured to receive and accumulate

the quantity of returnable packaging or pallets that is returned from the store over time and to compare the quantity returnable packaging or pallets that is returned from the store to a quantity of returnable packaging or pallets that is delivered to the store to determine a rate of asset loss.
